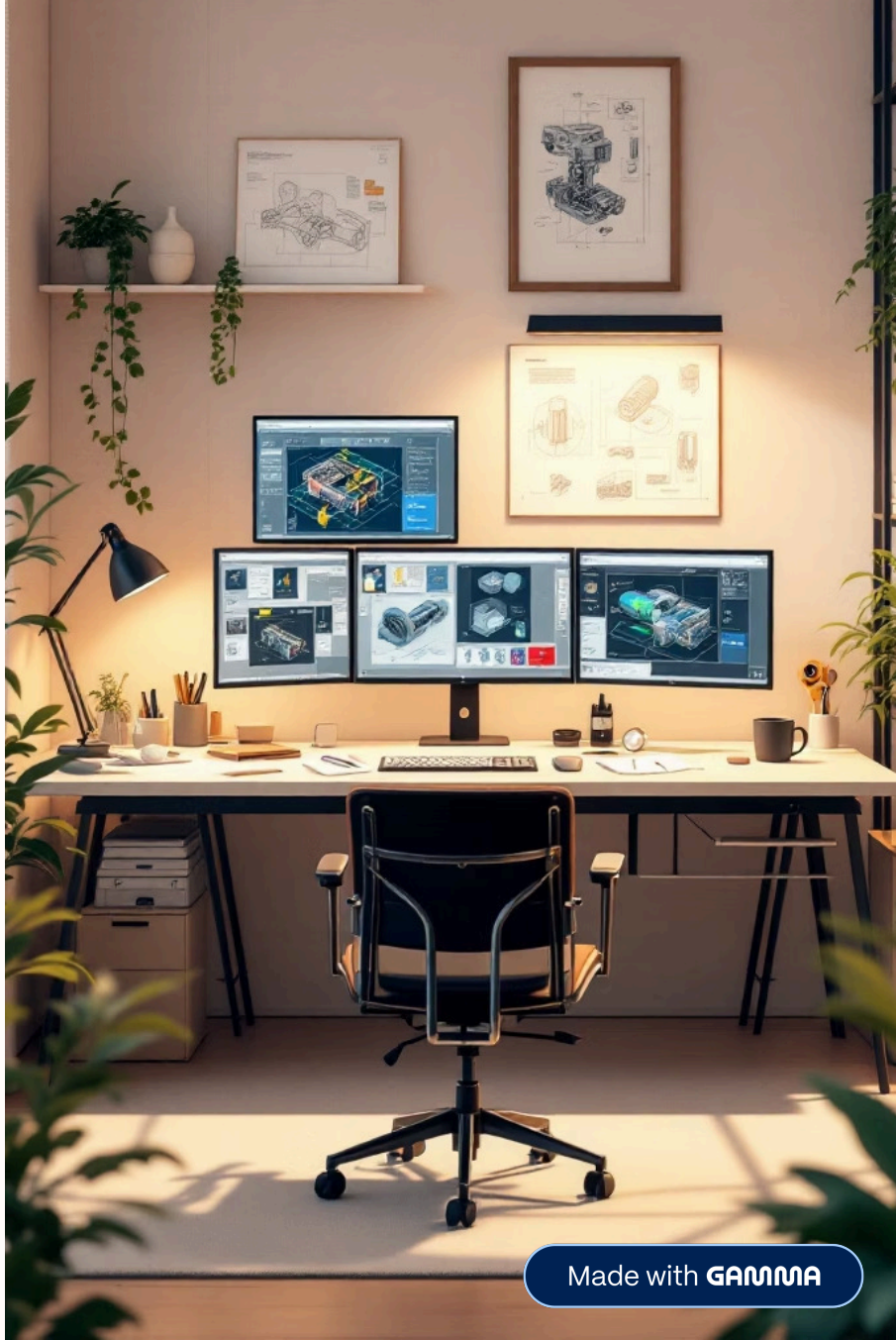
Project Run Guide

Three ways to execute the trolley design workflow

in local code editor in your computer



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Quick Start: Setup

01

Create Virtual Environment

python -m venv .venv

02

Activate Environment

.venv\Scripts\Activate.ps1

03

Upgrade pip

python -m pip install --upgrade pip

04

Install Dependencies

pip install -r requirements.txt

If PowerShell blocks activation, run: Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

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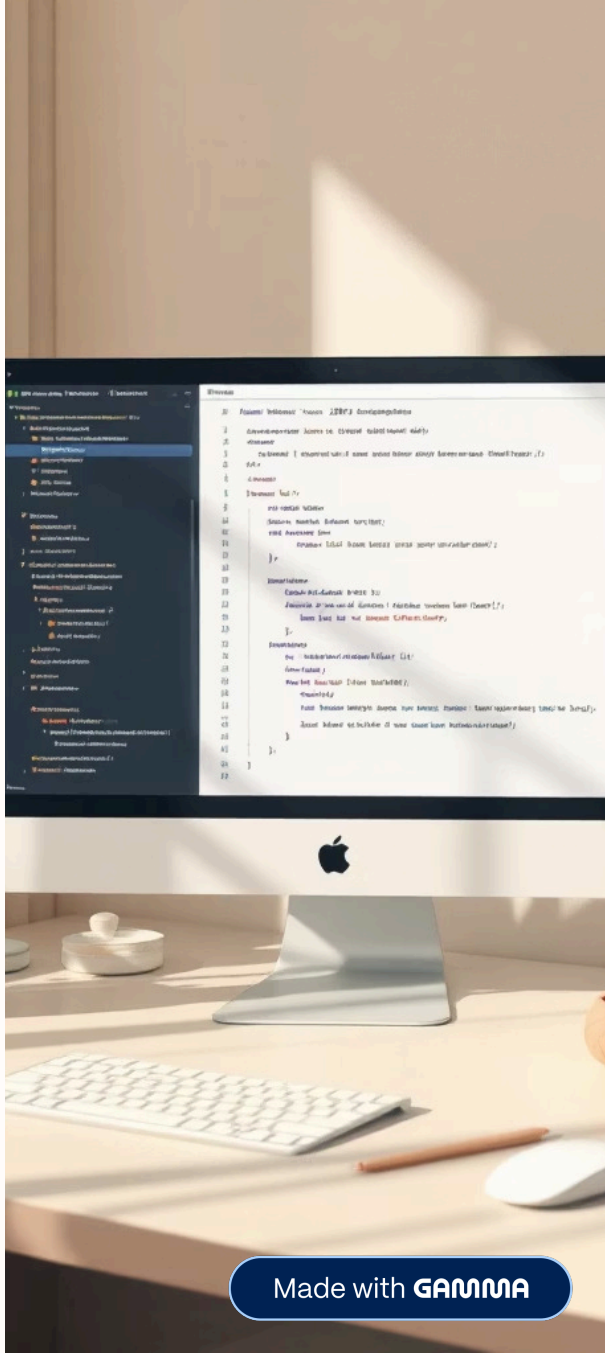
Method 1: JSON Input

Command

```
c:/automation/.venv/Scripts/python.exe src/main.py --input examples/sample_input_4.json --out outputs/run_sample_input_4
```

Generated Files

- summary.json
- trolley.scad (CAD)
- trolley_views.dxf (Drawings)
- costing_sheet.csv



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Method 2: CLI Arguments

Pass values directly in terminal instead of using JSON file

```
python src/main.py --use-cli-input --load-capacity-kg 900 --platform-length-mm 1600 --platform-width-mm 1000 --desired-material "Stainless Steel" --number-of-wheels 4 --operating-environment Indoor --out outputs/run_sample_input_4_cli
```

Same Output

Generates identical files to JSON method

Direct Control

No configuration file needed

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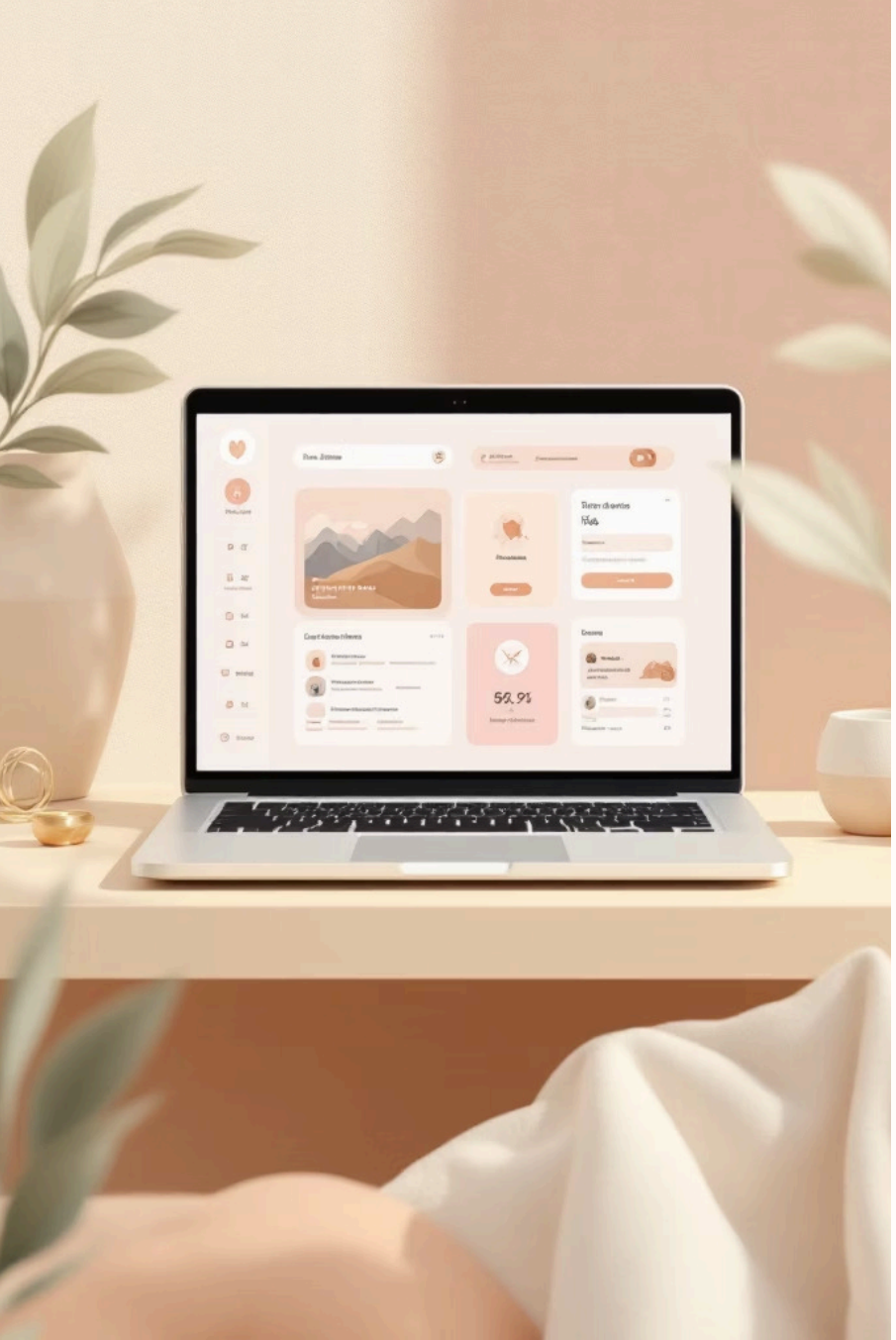

Method 3: Streamlit UI

1

Launch App

run " streamlit run frontend/app.py" on terminal

2

Open URL

http://localhost:8501

3

Use Form

Enter inputs and run

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